Daniel J Noh — (he/him)

Ph.D. Student, University of Pennsylvania

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danieljnoh.com | Google Scholar

Research Overview

learning sciences, care, maintenance & repair, participatory design, design education, computing education, child-computer interaction, responsible AI, constructionism, accessibility

Education

Expected 2029 | **University of Pennsylvania**
Ph.D., Learning Sciences & Technologies
Advisor: Yasmin B. Kafai

2022 | **Harvard University**
Ed.M., Learning Design, Innovation & Technology

2021 | **Carnegie Mellon University**
B.Arch., Architectural & Building Sciences/Technology (University & College Honors)
Minors: Human-Computer Interaction & Design for Learning

Research Experience

Aug 2024 to current | **University of Pennsylvania**
Graduate School of Education — Kafai Lab (PI: Yasmin Kafai)

NSF Grants #2333469, #2342438, #2414590
Spencer, Kapor, William T. Grant, Alfred P. Sloan Foundations Rapid Response Funding
Google Academic Research Award: AI for Privacy, Safety, & Security (2025)

Research Fellow
- Collecting and analyzing data to investigate learners' and educators' conceptions of AI/ML systems (e.g., generative language models, social media, and targeted advertisements) and algorithmic justice in formal and informal learning environments
- Developing related learning tools and activities through participatory design with educators and youth

Jun 2024 to Aug 2024 | **Northeastern University**
College of Arts, Media and Design (PI: Sara Hendren)

Research Assistant
- Developed experimental pedagogical modules for the studio-seminar course, *Investigating Normal: Design and Disability*, identifying best practices for accessible spatial and digital design, related precedent projects, and assessment strategies for proposed activities and curricular prompts

Sep 2021 to Jun 2022 | **Harvard University**
Project Zero — Designing Learning Places Lab (PI: Daniel Wilson)

Research Assistant
- Conducted a literature review on the qualities of materials, objects, and environments that support the learning practices of noticing, wondering, and help-seeking
- Designed a practical toolkit for practitioners to adopt the findings into the design of learning spaces
- Co-developed the course, *Designing Learning Places*, currently taught at the Harvard Graduate School of Education

Sep 2020 to Sep 2021 | **Carnegie Mellon University**
Human-Computer Interaction Institute — ClassInSight (PI: Amy Ogan)

James S. McDonnell Foundation Grant #1822813

Research Associate (May 2021 to Sep 2021)
Research Assistant (Sep 2020 to May 2021)
- Drafted and conducted focus group protocols, interview questions, and participatory design sessions to understand how classroom discussion data visualizations and personalized professional development can scaffold teacher agency and meaning-making
- Designed prototypes and data visualizations for a conversation support professional development tool

Honors & Awards

2026 | **Best Full Paper Award,** ACM IDC 2026
2024-2028 | **Fontaine Society Fellowship,** University of Pennsylvania
2021 | **GEE! Learning Games Award Finalist,** GEE! Learning Games Awards
2020 | **Louis F. Valentour Scholarship,** CMU School of Architecture

Publications

Journal Articles (Peer-Reviewed)

[J.1] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. B. (2025). High school students building babyGPTs: Engaging in data practices and addressing ethical issues through the construction of generative language models. *International Journal of Child-Computer Interaction, 45*, 100769.
DOI: https://doi.org/10.1016/j.ijcci.2025.100769

Conference Proceedings (Peer-Reviewed, Full)

[F.4] Morales-Navarro, L., **Noh, D. J.**, Servat, L., Netting, C., Kafai, Y. B., & Metaxa, D. (2026). Building to Understand: Examining Teens' Technical and Socio-Ethical Pieces of Understanding in the Construction of Small Generative Language Models. In *Proceedings of the 25th Interaction Design and Children,* 262–280.
DOI: https://doi.org/10.1145/3773077.3806107 [Acceptance Rate: 23%]
🏆 **Best Full Paper Award**

[F.3] **Noh, D. J.**, Fields, D. A., Kafai, Y. B., & Metaxa, D. (2026). "You can actually do something": Shifts in high school computer science teachers' conceptions of AI/ML systems and algorithmic justice. In Litts, B. K., DeLiema, D., Lee, C., Krist, S. Mawasi, A., Martin, L., & Kumpulainen, K. (Eds.), *Proceedings of the 20th International Conference of the Learning Sciences* (pp. 262-271). International Society of the Learning Sciences.
DOI: https://doi.org/10.22318/icls2026.716535 [Acceptance Rate: 30%]

[F.2] **Noh, D. J.** (2026). Guide on the Side or Sage on the Stage?: Exploring the Relationship Between Teachers' Spatial and Verbal Discursive Strategies. In G. Carmona, C. Lima, M. J. Santos, H. Benítez, L. Montero-Moguel, & B. Galarza-Tohen (Eds.), *Advances in Quantitative Ethnography* (Vol. 2677, pp. 509–523). Springer Nature Switzerland.
DOI: https://doi.org/10.1007/978-3-032-12229-2_33

[F.1] **Noh, D. J.**, Fields, D. A., Morales-Navarro, L., Cabrera-Sutch, A. M., Kafai, Y. B., & Metaxa, D. (2025). Youth as Advisors in Participatory Design: Situating Teens' Expertise in Everyday Algorithm Auditing with Teachers and Researchers. In *Proceedings of the 24th Interaction Design and Children*, 415–428.
DOI: https://doi.org/10.1145/3713043.3728849 [Acceptance Rate: 29%]

Conference Proceedings (Peer-Reviewed, Short & Symposia)

[S.5] Morales-Navarro, L., Fields, D., Giang, M. T., **Noh, D. J.**, Kafai, Y. & Metaxa, D. (2026). Understanding teens' self-beliefs when learning to construct and deconstruct AI/ML systems: Developing a survey instrument. In *Proceedings of the 25th Interaction Design and Children,* 1198–1202*.*
DOI: https://doi.org/10.1145/3773077.3812156

[S.4] Morales-Navarro, L., Kafai, Y. B., Hilke, O., Pope, N., Tedre, M., **Noh, D. J.**, Fields, D. A., Metaxa, D., Solyst, J., Shapiro, R. B., Fong, C., Nurdin, F., Sharma, S., Iivari, N., Klemettilä, P. A., Bilstrup, K. K., Musaeus, L. H., Landesman, R., Everson, J., Ryoo, J., & Iversen, O. S. (2026). Learning to deconstruct AI/ML systems: Scaffolds to support learners in critically evaluating AI/ML systems. In Litts, B. K., DeLiema, D., Lee, C., Krist, S. Mawasi, A., Martin, L., & Kumpulainen, K. (Eds.), *Proceedings of the 20th International Conference of the Learning Sciences*, (pp. 2204-2214). International Society of the Learning Sciences.
DOI: https://doi.org/10.22318/icls2026.998977

[S.3] Ching, C. C., Cortez, A., Higgs, J. M., Grover, S., Vakhil, S., Paik, W., Abdul, S., Knight, M., Lizárraga, J., Ryoo, J., Choi, M., Wei, W., Clarke, R., Blizzard-Caron, J., Stark, L., Kohn, L., Voloch, D., Ochterski, J., Wang, X. C., Xing, G. Y., Dantu, K., Morales-Navarro, L., **Noh, D. J.**, Kafai, Y. B., & Stornaiuolo, A. (2026). Artificial intelligence and human purposes: Dialogues about creativity and control. In Litts, B. K., DeLiema, D., Lee, C., Krist, S. Mawasi, A., Martin, L., & Kumpulainen, K. (Eds.), *Proceedings of the 20th International Conference of the Learning Sciences,* (pp. 2263-2274). International Society of the Learning Sciences.
DOI: https://doi.org/10.22318/icls2026.216764

[S.2] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. (2025). Building babyGPTs: Youth engaging in data practices and ethical considerations through the construction of generative language models. In *Proceedings of the 24th Interaction Design and Children*, 1021–1026.
DOI: https://doi.org/10.1145/3713043.3731525

[S.1] Kafai, Y., Shapiro, R. B., Jetzinger, F., Michaeli, T., Tedre, M., Vartiainen, H., Iivari, N., Musaeus, L. H., Iversen, O. S., Ali, S., Bodon, H., Butler, M., Kshirsagar, K., Smith, M., Quiterio, A., Worsley, M., Kumar, V., Morales-Navarro, L., **Noh, D.**, Pea, R., & Philip, T. M. (2025). Youth as Designers of Artificial Intelligence and Machine Learning Technologies: What do we Know About the Opportunities and Challenges of K-12 Students Creating Their Own Applications?. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025 (pp. 2260-2268). International Society of the Learning Sciences.
DOI: https://doi.org/10.22318/icls2025.362276

Posters, Presentations & Workshops (Peer-Reviewed)

[P.7] Porter, B., Okker-Edging, K., Liu, A., **Noh, D. J.**, Lydic, K., Muzekari, B., Lewis, T., Torres-Grillo, O., Falk, E., Tan, A. S.L. & Cosme, D. (2026, May). Youth Participatory Action Communication Research on Social and Environmental Determinants of Health. *2026 Conference for Advancing Participatory Sciences.* Virtual.

[P.6] **Noh, D.,** Fields, D. A., Kafai, Y. B. & Metaxa, D. (2026, April). "A Path to Activism": CS Teachers Developing Critical Agency with Algorithmic Systems through Algorithm Auditing. *American Educational Research Association (AERA) Conference 2026.* Los Angeles, CA.
DOI: https://doi.org/10.3102/2281161

[P.5] Morales-Navarro, L., **Noh, D.,** & Kafai, Y. B. (2026, April). High School Students as Designers of Generative Language Models. *American Educational Research Association (AERA) Conference 2026.* Los Angeles, CA.
DOI: https://doi.org/10.3102/2285159

[P.4] Morales-Navarro, L., **Noh, D.,** & Kafai, Y. B. (2026, April). Teens' Ethical Considerations When Building babyGPTs. *American Educational Research Association (AERA) Conference 2026.* Los Angeles, CA.
DOI: https://doi.org/10.3102/ip.26.2279543

[P.3] Fields, D. A., Morales-Navarro, L., **Noh, D.**, Kafai, Y. B. & Ottina, J. (2025, October). Workshop: Investigating the Black Box of Artificial Intelligence and Machine Learning: Promoting Youth and Teacher Agency through Algorithm Auditing. *Connected Learning Summit 2025.* Virtual.

[P.2] **Noh, D. J.**, Morales-Navarro, L., & Kafai, Y. (2025, October). What Comes Next?: Youth Learning About and Designing Markov Chains through Unplugged Activities. *Connected Learning Summit 2025.* Virtual.

[P.1] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. (2025, October). What happens when teens design small generative language models? A case study of teenagers building babyGPTs. *Connected Learning Summit 2025.* Virtual.

Invited Talks & Presentations

[T.2] **Noh, D. J.** (2026, June). What's Behind AI? Artificial Intelligence & Bias. Invited presentation and workshop for *The Academy At Penn.* Philadelphia, PA.

[T.1] **Noh, D. J.** (2026, April). Youth as Advisors and Teachers as Designers: The Participatory Design of AI Auditing Lessons. Invited presentation and workshop for *Penn GSE AI & Education Symposium.* Philadelphia, PA.

Public Scholarship (White Papers, Curriculum & Other)

[PS.2] Butapetch, H., Fields, D., Hanna, S., Landa. J., Morales-Navarro, L., Myers, T., **Noh, D.**, Ottina, J., & Ulrich, A., with Kafai, Y. & Metaxa. D. (2025). Introduction to Algorithm Auditing: Lessons for High School Computer Science (beta version).
Available at: https://bit.ly/k12aiaudit-lessons

[PS.1] Gonzalez, P. G., **Noh, D.**, & Wilson, D. (2022). Making the Space for Learning. *Presidents and Fellows of Harvard College. Cambridge, MA.*
Available at: https://pz.harvard.edu/resources/making-the-space-for-learning

Selected News and Press

[N.2] ***CMU IDeATe News.*** Alumni Spotlight: Daniel Noh.
https://ideate.cmu.edu/news-and-announcements/2026/08_august/alumni-spotlight-daniel.html

[N.1] ***MIT MAD News.*** Bringing Design Education to K-12 Students. (2023, April).
https://design.mit.edu/news/bringing-design-education-to-k-12-students

Teaching

Sep 2026 to current

University of Pennsylvania
Graduate School of Education

Co-Instructor, **MaKer Studio**
Taught with Yasmin Kafai (Spring 2027)

Co-Instructor, **AI for Children and Youth: Learning and Creating in K-12 Education**
Taught with Yasmin Kafai (Fall 2026)

Jan 2022 to Jan 2025

Harvard University
Graduate School of Education
**As a Teaching Fellow I was involved in course redesign, development of teaching material, leading weekly in-class lectures/workshops, and providing constructive student feedback*

Teaching Fellow, **Rapid Prototyping of Educational Products**
Instructor: Bertrand Schneider (Winter 2025)

Course Co-Designer & Lead Teaching Fellow, **Designing Learning Places**
Instructor: Daniel Wilson (Spring 2023, Spring 2024)

Teaching Fellow, **Digital Fabrication and Making in Education**
Instructor: Bertrand Schneider (Spring 2022, Fall 2022, Spring 2024)

Teaching Fellow, **Transforming Education Through Emerging Technologies**
Instructor: Bertrand Schneider (Spring 2024)

Teaching Fellow, **Multimodal Learning Analytics**
Instructor: Bertrand Schneider (Spring 2023)

Feb 2024 to May 2024

Design Museum Everywhere
High School Inclusive Design Challenge (Co-Instructor: Rosa Weinberg)

Design Educator
- Planned and facilitated a three-week long program on accessibility, inclusive design, and participatory design with six financially-compensated high school students

Sep 2022 to Jan 2024

Massachusetts Institute of Technology
MIT Museum

Technical Instructor / Museum Educator
- Designed and facilitated novel hands-on STEAM learning experiences and public events to build meaningful connections to scientific inquiry and design exploration
- Established standard operating procedures and trained museum educators on various fabrication tools in the makerspace (laser cutter, 3D printers, and vinyl cutters)
- Mentored MIT undergraduate research assistants (UROP) and museum teen interns on the development and facilitation of design-based, hands-on learning experiences
- Co-led the launch and development of the Design Redefined series, co-hosted by MIT MAD and Innovators for Purpose

Advising & Mentoring

^ = Indicates co-authorship in publication

Undergraduate Researchers

2026-2027 - Sophia Wang (University of Pennsylvania)
2025-2026 - Lucianne Servat^ (University of Pennsylvania, with Luis Morales-Navarro)
2025 - Elo Esalomi (University of Pennsylvania, with Luis Morales-Navarro)
2023-2024 - Wonuola Abiodun (MIT, with Rosa Weinberg → **now Stanford PhD**)
2023-2024 - Alexandra Coston (MIT, with Rosa Weinberg)
2023 - Kimberly McPherson (MIT, with Rosa Weinberg)

Service

Journal Reviewing: Journal of the Learning Sciences (2026)

Conference Reviewing:
ACM Interaction Design & Children (2026)
International Conference of the Learning Sciences (2026)
Symposium on Educational Advances in Artificial Intelligence (2025-2026)
Connected Learning Summit (2025)
International Conference on Quantitative Ethnography (2025)

Institutional Service

Massachusetts Institute of Technology
- MIT-Nord Anglia, STEAM Teacher Professional Development Mentor (2024)

Carnegie Mellon University
- School of Architecture, Student Portfolio Reviewer (2021)
- School of Architecture, Architecture Peer Mentor (2017)

Other Service

Carnegie Museum of Natural History
- Museum Volunteer (2020)

Other Experience

Aug 2021 to May 2022

Harvard University
Graduate School of Education – Education Innovation Studio

Makerspace Assistant
- Taught graduate students how to use fabrication tools including laser cutters, 3D printers, vinyl cutters, sewing machines, and power tools

May 2020 to Aug 2020

Carnegie Museum of Natural History
Center for Anthropocene Studies (Advisors: Asia Ward & Marti Louw)

Center for Anthropocene Studies Education Intern
- Produced educational videos about sewage treatment and potable water, presented at ALCOSAN's 2020 and 2021 Open House
- Published corresponding articles in the museum's monthly newsletters and website

May 2018 to Jan 2021

Carnegie Mellon University
School of Architecture

Design Fabrication (dFab) Lab Assistant
- Assisted architecture students and faculty on implementing digital fabrication methods (laser cutters, 3D printers, CNC-mill, and vacuum formers) into their project workflow

Feb 2019 to May 2020

Carnegie Mellon University
School of Architecture, Manufacturing Futures Initiative (PI: Joshua Bard)

Research Assistant
- Investigated the use of steam-bending and novel joinery to design, prototype, and fabricate steam-bent, wooden swings for a public park in Pittsburgh

Skills & Tools

Programming | Python, JavaScript, HTML/CSS, R

Fabrication | Laser Cutting, 3D Printing, Vinyl Cutting, Vacuum Forming, Sewing, Woodworking, Soldering, Basic Electronics/Microcontrollers

Design | Rapid Prototyping, Wireframing, Storyboarding, Illustrating, Data Visualization, Graphic Design, Architectural Design, Learning Design, Video Editing, Web Design & Development

Methods | Thematic Analysis, Interviews, Participatory Design Research, Youth Participatory Action Research, Multi-Modal Learning Analytics, Epistemic/Ordered Network Analysis

Professional Memberships

American Education Research Association (AERA) – DIV C
International Society of the Learning Sciences (ISLS)
Association of Computer Machinery (ACM) – SIGCHI